

◆ AWS Solutions Architect Associate (SAA-C03)

◆ Learning Notes

Compute Security, Reserved Instances, Spot Instances & Cost Optimization

◆ Learning Objectives

By the end of this lesson, you should be able to:

- ✓ Understand AWS Compute Security
 - ✓ Explain Security Groups
 - ✓ Understand IAM Roles for EC2
 - ✓ Differentiate On-Demand, Reserved, and Spot Instances
 - ✓ Select the correct pricing model for workloads
 - ✓ Apply AWS Cost Optimization principles
 - ✓ Analyze real SAA exam scenarios
 - ✓ Make architecture decisions based on cost, security, and scalability
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◆ 1. Compute Security Fundamentals

What Is Compute Security?

Compute Security refers to protecting compute resources such as:

- Amazon EC2
- Auto Scaling Groups
- Containers

- Serverless workloads

from unauthorized access, attacks, and data breaches.

Why Compute Security Matters

A compromised server can lead to:

- ✗ Data theft
 - ✗ Service disruption
 - ✗ Financial losses
 - ✗ Compliance violations
 - ✗ Reputational damage
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AWS Shared Responsibility Model

AWS Secures

- ✓ Physical Data Centers
 - ✓ Hardware
 - ✓ Networking Infrastructure
 - ✓ Hypervisor
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Customer Secures

- ✓ Operating Systems
- ✓ Applications
- ✓ User Access
- ✓ Security Groups

✓ IAM Permissions

✓ Data

SAA Exam Tip

AWS secures the cloud.

Customers secure what they place in the cloud.

◆ 2. Security Groups

What Is a Security Group?

A Security Group is a virtual firewall that controls traffic entering and leaving an EC2 instance.

Characteristics

Stateful

If inbound traffic is allowed:

Response traffic is automatically allowed.

Instance-Level Protection

Security Groups are attached directly to resources.

Examples:

- EC2
 - RDS
 - Load Balancers
-

Allow Rules Only

Security Groups contain:

✓ Allow Rules

They do NOT contain:

✗ Deny Rules

Example

Web Server

Allow:

HTTP (80)

HTTPS (443)

Deny:

All Other Traffic

SAA Exam Tip

When the exam asks:

Which AWS service acts as a virtual firewall?

Answer:

✓ Security Group

◆ 3. Network ACL vs Security Group

Students commonly confuse these concepts.

Security Group

Feature	Security Group
Stateful	Yes

Feature	Security Group
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Allow Rules	Yes
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Deny Rules	No
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Attached To	Instance
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Network ACL

Feature	Network ACL
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Stateful	No
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Allow Rules	Yes
-------------	-----

Deny Rules	Yes
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Attached To	Subnet
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Exam Memory Trick

Security Group
=
Instance Firewall

NACL
=
Subnet Firewall

◆ 4. IAM Roles for EC2

Problem

An EC2 instance needs access to S3.

Many beginners attempt to:

✗ Store AWS Access Keys on the server.

Why This Is Dangerous

Keys can:

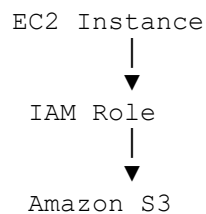
- Be stolen
- Expire
- Leak to GitHub
- Create security risks

AWS Best Practice

Use:

✓ IAM Roles

How It Works



Benefits

- ✓ Temporary Credentials
 - ✓ Automatic Rotation
 - ✓ Improved Security
 - ✓ Least Privilege Access
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SAA Exam Tip

If EC2 requires access to AWS services:

Answer:

✓ IAM Role

NOT

✗ Access Keys

◆ 5. EC2 Pricing Models

AWS provides several purchasing options.

Choosing the correct option can dramatically reduce costs.

◆ 6. On-Demand Instances

Definition

Pay only for what you use.

No commitment required.

Advantages

- ✓ Flexible
 - ✓ No upfront payment
 - ✓ No long-term commitment
-

Best For

- Testing
 - Development
 - Short-term projects
 - Unpredictable workloads
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Example

A startup launches a new application.

Traffic patterns are unknown.

Recommended:

✓ On-Demand Instances

Exam Tip

If the question says:

Unknown demand

Choose:

✓ On-Demand

◆ 7. Reserved Instances (RI)

Definition

Commit to AWS usage for:

- 1 Year
- 3 Years

in exchange for discounted pricing.

Benefits

Potential savings can be substantial compared to On-Demand pricing.

Best For

✓ Databases

✓ ERP Systems

✓ Production Applications

✓ Long-Term Workloads

Example

A company runs:

24/7 Production Database

for several years.

Recommended:

✓ Reserved Instance

Why?

Because usage is predictable.

SAA Exam Tip

Keywords:

- Predictable workload
- Long-term usage
- Cost savings

Answer:

✓ Reserved Instance

◆ 8. Spot Instances

Definition

Purchase unused AWS compute capacity at steep discounts.

Benefit

Can be significantly cheaper than On-Demand pricing.

Trade-Off

AWS may terminate the instance with short notice.

Best For

- ✓ Batch Processing
 - ✓ Big Data
 - ✓ CI/CD
 - ✓ Image Rendering
 - ✓ Machine Learning Training
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Not Suitable For

- ✗ Critical Databases
 - ✗ Transaction Systems
 - ✗ Sensitive Production Workloads
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Example

A company processes reports overnight.

If interrupted:

The job can restart.

Recommended:

- ✓ Spot Instances
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Exam Tip

Keywords:

- Interruptible
- Flexible
- Temporary
- Batch Job

Answer:

✓ Spot Instance

◆ 9. Savings Plans

What Are Savings Plans?

Flexible pricing model that provides discounts in exchange for a usage commitment.

Benefits

- ✓ Lower Costs
 - ✓ More Flexibility
 - ✓ Easier Management
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Exam Tip

AWS increasingly recommends Savings Plans for flexible cost optimization.

◆ 10. Cost Optimization

What Is Cost Optimization?

Using AWS resources efficiently while minimizing unnecessary spending.

AWS Well-Architected Pillar

Cost Optimization is one of the pillars of the:

AWS Well-Architected Framework

◆ Cost Optimization Strategy 1

Right-Sizing

Choose the correct instance size.

Poor Example

`m5.4xlarge`

Running at:

5% CPU

Better Option

`t3.medium`

Result

Lower monthly costs.

◆ Cost Optimization Strategy 2

Auto Scaling

Automatically add resources during demand spikes.

Automatically remove resources during low demand.

Benefits

- ✓ High Availability
 - ✓ Lower Costs
 - ✓ Improved Efficiency
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◆ Cost Optimization Strategy 3

Use Reserved Instances

For predictable workloads.

Examples:

- Databases
 - Internal Applications
 - ERP Systems
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◆ Cost Optimization Strategy 4

Use Spot Instances

For flexible workloads.

Examples:

- Testing
 - Batch Jobs
 - Analytics
-

◆ Cost Optimization Strategy 5

Monitor Costs

Use:

AWS Cost Explorer

Analyze cloud spending.

AWS Budgets

Set spending limits and alerts.

AWS Trusted Advisor

Receive cost optimization recommendations.

◆ Real SAA Exam Scenario

Scenario

A company operates:

Production Environment

Runs continuously.

Nightly Batch Processing

Runs for 4 hours every night.

Requirement

Reduce AWS Costs.

Best Architecture

Production Servers

✓ Reserved Instances

Batch Processing

✓ Spot Instances

Scaling

✓ Auto Scaling Group

Why?

Provides:

✓ Cost Optimization

✓ Reliability

✓ Scalability

✓ AWS Best Practices

◆ SAA Exam Questions

Question 1

An EC2 instance requires access to S3.

What should be used?

A. Access Keys

B. Security Group

C. IAM Role

D. NACL

Answer

✓ C. IAM Role

Question 2

Which pricing option is best for a predictable workload running 24/7?

- A. Spot
- B. Reserved Instance
- C. On-Demand
- D. Lambda

Answer

✓ B. Reserved Instance

Question 3

Which pricing model offers the lowest cost but may be interrupted?

- A. Reserved
- B. On-Demand
- C. Spot
- D. Dedicated Host

Answer

✓ C. Spot

Question 4

Which AWS service acts as a virtual firewall?

- A. IAM
- B. Security Group
- C. Route 53
- D. CloudTrail

Answer

✓ B. Security Group

Question 5

Which service helps analyze AWS spending?

- A. CloudFront
- B. EC2
- C. Cost Explorer
- D. Route 53

Answer

✓ C. Cost Explorer

◆ Quick Exam Memory Tricks

Compute Security

Security Group
=
Instance Firewall

IAM Access

EC2 → IAM Role → AWS Services

Pricing Models

On-Demand
=
Flexible
Reserved
=
Predictable
Spot
=
Cheapest

Cost Optimization

Remember:

RASM

- **Right-Sizing**
 - **Auto Scaling**
 - **Spot Instances**
 - **Monitor Costs**
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◆ Key Takeaways

- ✓ Security Groups are stateful virtual firewalls
 - ✓ IAM Roles are the secure way for EC2 to access AWS services
 - ✓ On-Demand Instances are best for uncertain workloads
 - ✓ Reserved Instances are best for predictable workloads
 - ✓ Spot Instances provide the lowest cost for flexible workloads
 - ✓ Auto Scaling improves both availability and cost efficiency
 - ✓ Cost Explorer, Budgets, and Trusted Advisor help optimize spending
 - ✓ AWS architects are expected to balance security, performance, reliability, and cost
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AWS Reference Resources

- [Amazon EC2 Pricing Models](#)
- [EC2 Security Best Practices](#)
- [AWS Cost Management](#)
- [AWS Well-Architected Framework](#)

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